

FES 511: Ecological Foundations for Environmental Managers

Fall-2, 2023: October 23rd – December 8th

Instructor: Elizabeth Forbes

elizabeth.forbes@yale.edu

Office hours: Tuesdays, 9:30-10:30am and 12pm-1pm (Kroon G17 and on [Zoom](#))

TFs:

- Mizzy Almaazmi
 - mizna.almaazmi@yale.edu
 - **Office hours:** Mondays 12:15-1:15pm (Kroon G15 and on Zoom)
- Amanda Wik
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 - **Office hours:** Thursdays 12-1PM (Kroon G19 and on [Zoom](#))

Course website:

<https://yale.instructure.com/courses/91141>

Course description

A fundamental environmental concern is whether natural ecosystems are resilient to natural or human-induced disturbances and sustain their functions. Ecology, as a scientific discipline, offers key theory, tools and empirical insights that can help answer important questions about ecosystem sustainability. This course will provide students with the necessary concepts and tools to begin evaluating how ecological systems are structured and how they respond to natural and human-induced perturbations.

Important dates:

- Our final exam will take place during finals week in the window of Dec. 15th – 20th. I will update the syllabus as soon as final exam dates are assigned for each course!
- No class during November recess (Nov. 17th – Nov. 27th)
- In-class mid-course evaluation: Thursday Nov. 9th, 11:40-11:50pm (please bring your laptop!)
- *****Weekly blog posts due every Sunday at 11:59pm!!*****
- ***Mid-course assignment: Op-ed due Sunday, Nov. 19th at 11:59pm***

Course assignments, grade breakdowns: all assignments will have associated rubrics

- Blog-style synopsis of papers and discussions (50%)
 - Once per week you will submit a 600-word written piece in the style of a blog post, synthesizing the one of your weekly readings, and explaining how and/or why (in your viewpoint and interpretation) the results are important for environmental management, conservation, justice, etc. Each blog will be worth 10 points (we'll skip the last week, so five blogs total). **Blogs for each week will be due by Sunday at 11:59pm (e.g. after the week is over).**
 - **UPDATE WEEK OF NOV. 20TH: the final blog post will be optional. Those who do not complete it will have their final grade out of 90 points; those who do complete it will have their final grade out of 100 points.**
 - These blogs will showcase your progression in several arenas: reading and synthesizing ecology publications from research journals; explaining those studies succinctly and effectively in short-form writing pieces to a general audience; interpreting those studies in

- your own words in terms of a relevant conservation, policy, justice, or related environmental challenge
 - These are skills that you'll need in your careers as environmental managers!
- OpEd essay (25%)
 - This assignment will require you to use your skills from blog-post writing (interpreting peer-reviewed research, forming your own opinions and ideas after synthesizing the ideas in the research paper, translating that synthesis and your ideas into straightforward prose for an imaginary interested audience).
 - However, it will also require you to remove yourself one step further from the scientific research being done
 - What are this author's goals? Does the author accomplish their goals?
 - Do you agree with their conclusions?
 - What are they missing, if anything?
 - And finally, **what is your own opinion of the subject (or, a related subject their piece contributes information to)? See "what is an op-ed"!**
 - We're not looking for a book report-style summary! We want critical thinking and your own voice to come through.
- Participation (5%)
 - Participation is required; however, my view is that participation takes many forms, and folks' different learning styles are suited to different kinds. We'll have multiple opportunities for participation and engagement, including group discussions, breakout groups, polls, etc. You should have no problem getting full participation credit as long as you attend class, no matter your learning style.
- Final exam (20%)
 - The final exam will be a mix of short-answer and longer-form writing responses to prompts based on the content from this semester's coursework and the required readings.

Attendance

For any lecture or other meeting that you are not able to attend I ask that you provide me, Mizzy, or Amanda reasonable notice (i.e. 24hrs) ahead of time. You may miss one lecture with no explanation needed, but you will need to provide an explanation if more than one lecture needs to be missed. You will be responsible for covering the material from the classes you miss on your own, though the teaching team will be available in office hours for you. This course is short and intensive, and we are covering a lot in each class; my priority is for you to get as much out of it as you can, and having to catch up on 3hrs of lectures on the weekends is not an ideal use of your time. If you have concerns about attendance, do not hesitate to reach out to me (Elizabeth) to discuss them.

Extension policy

If you require an extension for an assignment, please email your TFs 24hrs or more ahead of a due date for the assignment in question.

Code of conduct

We will spend time in our first lecture establishing a classroom code of conduct and respect. I will ask that we all agree to abide by the standards we come up with together so we can make our classroom a welcoming place for everybody to learn. I will also check in with the class regularly throughout our six weeks together to make sure everybody's learning objectives are being met. Please take this as an invitation to speak to any member of the teaching team if you have specific questions about our classroom's code of conduct throughout Fall-2! Our doors are and will remain open to you.

Campus resources

- Dr. Eva Wilson, graduate school's mental health clinician ([link to bookings page here](#))
- Mental Health and Counseling [resources](#)
- [Yale Well](#)

Feedback and anonymous help: As your teaching team, we want to be able to check in with you. Lots of folks find the ability to be anonymous helpful, so we're providing [this anonymous form](#) to you so you can send us notes throughout the semester.

[Glossary of Terms](#): please feel free to contribute!

Course text: *This text is free for you to read with an account on ebookcentral*

O.J. Schmitz 2007. Ecology and Ecosystem Conservation. Island Press.

<https://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=3317639>

MOTIVATING COURSE QUESTIONS:

What is ecology, and why do we care about learning about it?

How is ecology research conducted, and with what methods?

How is ecology research applied to modern-day environmental challenges and crises?

How can environmental managers leverage knowledge about ecology in their work?

Week 1

Tuesday, October 24th

Course Introduction: What is ecology? Why should we care?

[Link to form to submit suggestions for course "Code of Conduct"](#)

Readings:

- *Text Chapter 1 and Chapter 2*
- Link to reading #1: ["Biodiversity conservation needs a more ecological context"](#). Mongabay, 17 Feb. 2023
- Link to reading #2: ["What's in a name? The role of defining 'wilderness' in conservation"](#). Mongabay, 7 Feb. 2019

Thursday, October 26th

The study of our home: Earth System Science

Readings:

- *Text Chapter 3*
 - W. Steffen et al. 2020. The emergence and evolution of Earth System Science. *Nature Reviews Earth & Environment* 1:54-63
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Week 2

Tuesday, October 31st

Biodiversity: what is biodiversity, and how do we quantify it?

Readings:

- Text Chapter 6
- Wilkinson et al. 2020. An ecological framework for contextualizing carnivore-livestock conflict. *Conservation Biology*.
 - Check out Christine's twitter page [here](#), and her Google scholar page [here](#)

Thursday, November 2nd

Causes of biodiversity: niches, interspecies interactions, and more

Readings:

- Text Chapter 7
 - Gaynor et al. 2019. Landscapes of fear: Spatial patterns of risk perception and response. *Trends in Ecology and Evolution* 34(4): 355-368
 - Check out Kaitlyn's lab webpage [here](#), and their twitter page [here](#)
 - PODCAST: Ologies: [The life and death of an LA cougar](#) with Miguel Ordeñana and Beth Pratt
 - Link to transcript [here](#)
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Week 3

Tuesday, November 7th

Biogeography: the global distribution of species

Readings:

- Windsor et al. 2022. Using ecological networks to answer questions in global biogeography and ecology. *Journal of Biogeography* DOI: 10.1111/jbi.14447
- PODCAST: Ologies: [Opossumology](#) with Dr. Lisa Walsh
 - Link to transcript [here](#)

Thursday, November 9th

Biodiversity and ecosystem functioning

Readings:

- Text chapter 8
 - Forbes et al. 2019. Synthesizing the effects of large, wild herbivore exclusion on ecosystem function. *Functional Ecology* 33: 1597-1610
 - ~~PODCAST: Ologies: [Carnivore ecology](#) with Dr. Rae Wynn-Grant ← no longer need to listen to this one~~
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Week 4

Tuesday, November 14

Ecosystem functioning: nutrient cycling

Readings:

- Ferraro, Schmitz, and McCary 2022. Effects of ungulate density and sociality on landscape heterogeneity: A mechanistic modeling approach. *Ecography* e06039. Doi: 10.1111/ecog.06039
- PODCAST: Ologies episode: [Indigenous Pedology](#) with Dr. Lydia Jennings

- Link to transcript [here](#)

Thursday, November 16

Ecosystem resilience, resistance, and change

Readings:

- Text chapter 9
- Zavaleta et al. 2009. Ecosystem responses to community disassembly. "The year in Ecology and Conservation Biology": *Ann. N.Y. Acad. Sci.* 1162: 311-333.
- PODCAST: Ologies episode: [Acaropathology](#) (ticks and Lyme disease) with Dr. Neeta Pardanani Gonnally, Dr. Andrea Swei ← **no longer need to listen to this one**

****OP-ED DUE NOVEMBER 19TH (SUNDAY), 11:59PM, PRIOR TO FALL BREAK****

Week 5

Tuesday, November 28

Ecosystem succession and challenges

Readings:

- Text chapter 10
- Leroux et al. 2021. Cumulative effects of spruce budworm and moose herbivory on boreal forest ecosystems. *Functional Ecology* 35: 1448-1459. DOI: 10.1111/1365-2435.13805
- PODCAST: Ologies episode: [Fire Ecology](#) with Dr. Gavin Jones, Dr. Amy Christianson
 - **Link to transcript [here](#)**

Thursday, November 30th

Conservation: its history and modernization

Readings:

- Text chapter 5
- 3x lobster fisheries articles:
 - #1: [legislation on fishing gear in regard to right whales](#)
 - #2: [legal size of lobster](#)
 - #3: [costs of using ropeless fishing gear](#)

Week 6

Tuesday, December 5th

Climate Change

Readings:

- Voigt et al. 2023. Arctic soil methane sink increases with drier conditions and higher ecosystem respiration. *Nature Climate Change* 13: 1095-1104
- PODCAST: Ologies: [Phenology](#) with Dr. Libby Ellwood
 - **Link to transcript [here](#)**

Thursday, December 7

Conservation and climate change: using tools from ecology to study and address modern environmental challenges

Readings:

- Cabral et al. 2020. A global network of marine protected areas for food. *PNAS* 117(45): 28134-28139